Product Gales and Finite State Dimension

Akhil S

Outline

1. Overview
2. Normality
3. Entropy
4. Automata and Entropy
5. Multi-bet Gales

Overview

- Finite-State dimension ($\dim_{\mathrm{FS}}$)
  - Measures Information Density
    - ⋆ Defined using Finite-state Gamblers (s-gales).
    - ⋆ Equivalently, using *disjoint* entropy rates ($H^{disjoint}$). [BHV05 [5]]
    - ⋆ $H^{disjoint} = H^{sliding}$. [KS21 [6]]

Overview

- Finite-State dimension ($\dim_{\mathrm{FS}}$)
  - Measures Information Density
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    - ⋆ Equivalently, using *disjoint* entropy rates ($H^{disjoint}$). [BHV05 [5]]
    - ⋆ $H^{disjoint} = H^{sliding}$. [KS21 [6]]
- We Introduce : **Multi-bet** Finite-State Dimension ($\dim_{\mathrm{FS}}^{\mathrm{multi-bet}}$).
  - Defined using k-**bet** Finite-state s-gales.

Overview

- We Introduce
  - Finite-state muliti-bet gales.
  - Multi-bet finite-state dimension ($\dim_{\mathrm{FS}}^{\mathrm{multi-bet}}$).
- We Show :

$$\dim_{\mathrm{FS}}^{\mathrm{multi-bet}} = H^{sliding}.$$
$$\dim_{\mathrm{FS}}^{\mathrm{multi-bet}} = H^{disjoint}.$$

Overview

- Give an alternate proof of (Shen and Kozachinsky [6]):

  Sliding block entropy rates = Disjoint block entropy rates .

$$H^{disjoint} = H^{sliding}.$$

- Using $\dim_{FS} = H^{disjoint}$ (BHV 2005 [5]).

$$\dim_{\mathrm{FS}}^{\mathrm{multi-bet}} = \dim_{\mathrm{FS}}.$$

Outline

1 Overview

2 Normality

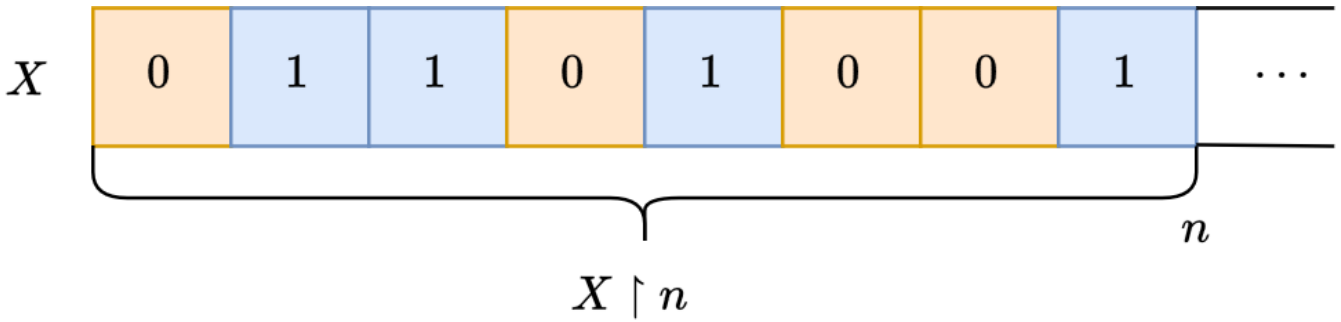
3 Entropy

4 Automata and Entropy

5 Multi-bet Gales

Normality

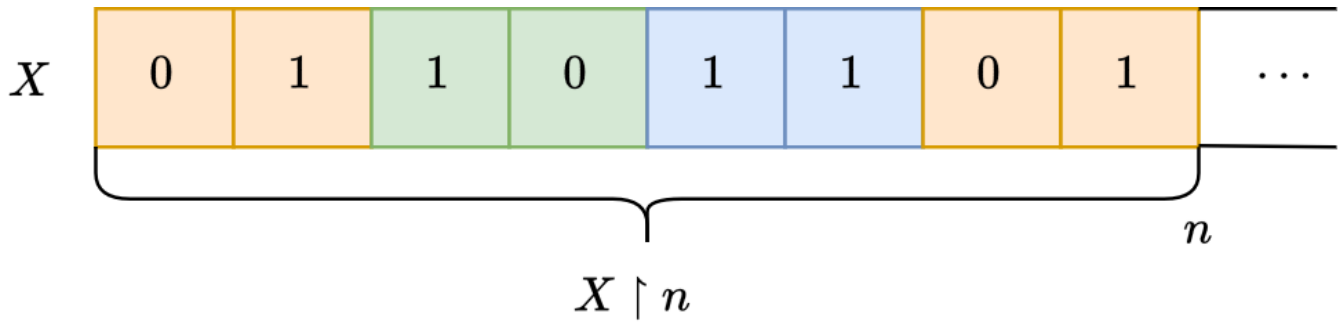
- A "random" binary string should have half 0's and half 1's in the limit.



$$\lim_{n\to\infty} \frac{\#_0(X \upharpoonright n)}{n} = \frac{1}{2} \qquad \lim_{n\to\infty} \frac{\#_1(X \upharpoonright n)}{n} = \frac{1}{2}$$

Normality

- A "random" binary string should have strings $00, 01, 10, 11$ in the same ratio $(1/4)$.



$$\lim_{n\to\infty} \frac{\#_{00}(X \upharpoonright n)}{n/2} = \lim_{n\to\infty} \frac{\#_{01}(X \upharpoonright n)}{n/2} = \lim_{n\to\infty} \frac{\#_{10}(X \upharpoonright n)}{n/2} = \lim_{n\to\infty} \frac{\#_{11}(X \upharpoonright n)}{n/2} = \frac{1}{4}$$

Normality

Definition

A sequence $X \in \Sigma^\infty$ is ℓ-normal iff $\forall w \in \Sigma^\ell$,

$$\lim_{k\to\infty} \frac{\#_w(X \upharpoonright k\ell)}{k} = 2^{-\ell}.$$

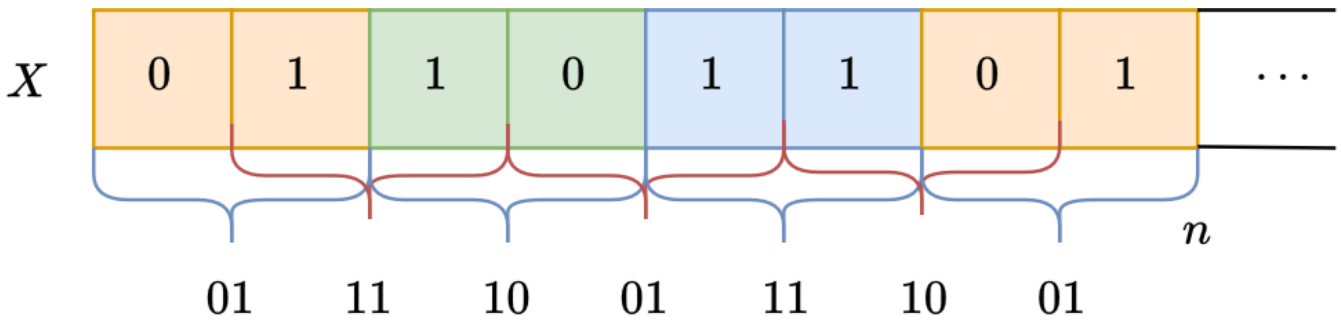
Definition

A sequence $X \in \Sigma^\infty$ is Normal iff $\forall \ell \in \mathbb{N}$, $\forall w \in \Sigma^\ell$,

$$\lim_{k\to\infty} \frac{\#_w(X \upharpoonright k\ell)}{k} = 2^{-\ell}.$$

Sliding Normality

- What if we count ℓ-length blocks in a *sliding* manner ?.



$$\lim_{n\to\infty} \frac{\#'_{00}(X \upharpoonright n)}{n-1}, \lim_{n\to\infty} \frac{\#'_{01}(X \upharpoonright n)}{n-1}, \lim_{n\to\infty} \frac{\#'_{10}(X \upharpoonright n)}{n-1}, \lim_{n\to\infty} \frac{\#'_{11}(X \upharpoonright n)}{n-1}$$

Sliding Normality

Definition

A sequence $X \in \Sigma^\infty$ is ℓ-sliding normal iff $\forall w \in \Sigma^\ell$,

$$\lim_{n\to\infty} \frac{\#'_w(X \upharpoonright n)}{n-\ell+1} = 2^{-\ell}.$$

Definition

A sequence $X \in \Sigma^\infty$ is Sliding Normal iff $\forall \ell \in \mathbb{N}$, $\forall w \in \Sigma^\ell$,

$$\lim_{n\to\infty} \frac{\#'_w(X \upharpoonright n)}{n-\ell+1} = 2^{-\ell}.$$

Pillai's Thoerem

Theorem (S. S. Pillai 1940 [2])

For any sequence $X \in \Sigma^{\infty}$,

$$X \text{ is Sliding Normal} \iff X \text{ is Disjoint Normal.}$$

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Figure: Subbayya Sivasankaranarayana Pillai

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Normality vs. Entropy

$$X \in \Sigma^\infty \text{ is Disjoint Normal } \iff H^{disjoint}(X) = 1.$$

Normality vs. Entropy

$$X \in \Sigma^{\infty} \text{ is Disjoint Normal } \iff H^{disjoint}(X) = 1.$$

$$X \in \Sigma^{\infty} \text{ is Sliding Normal } \iff H^{sliding}(X) = 1.$$

Shannon Entropy

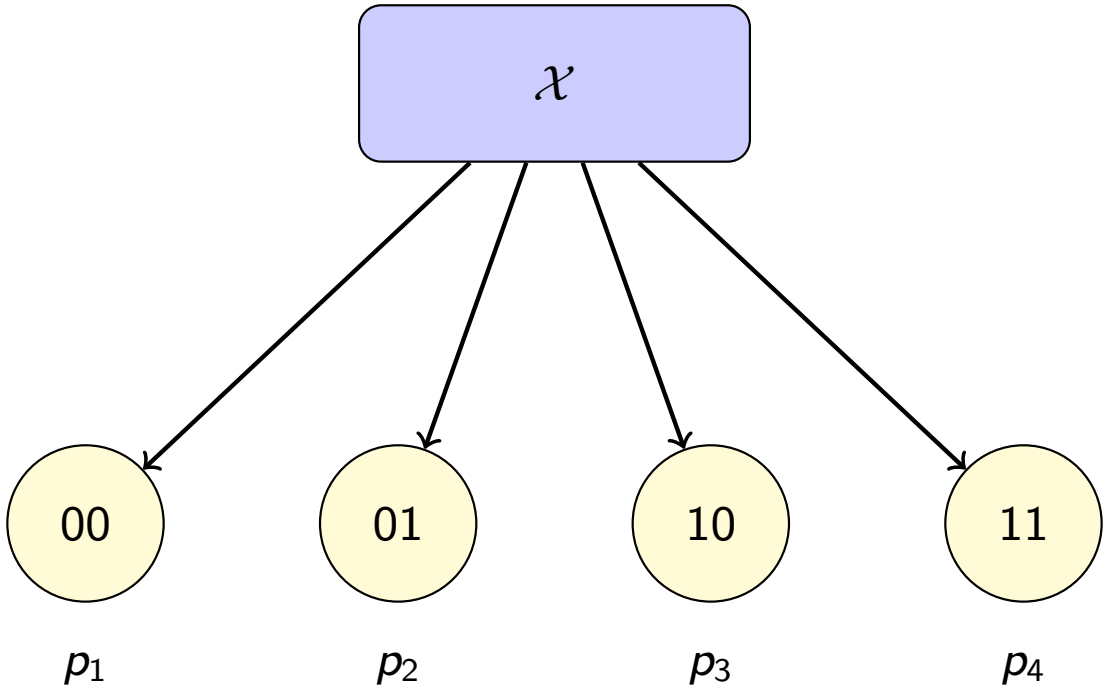
Definition

The *Shannon entropy* of a distribution $p(x)$ over a finite space $\mathcal{X}$ is defined as:

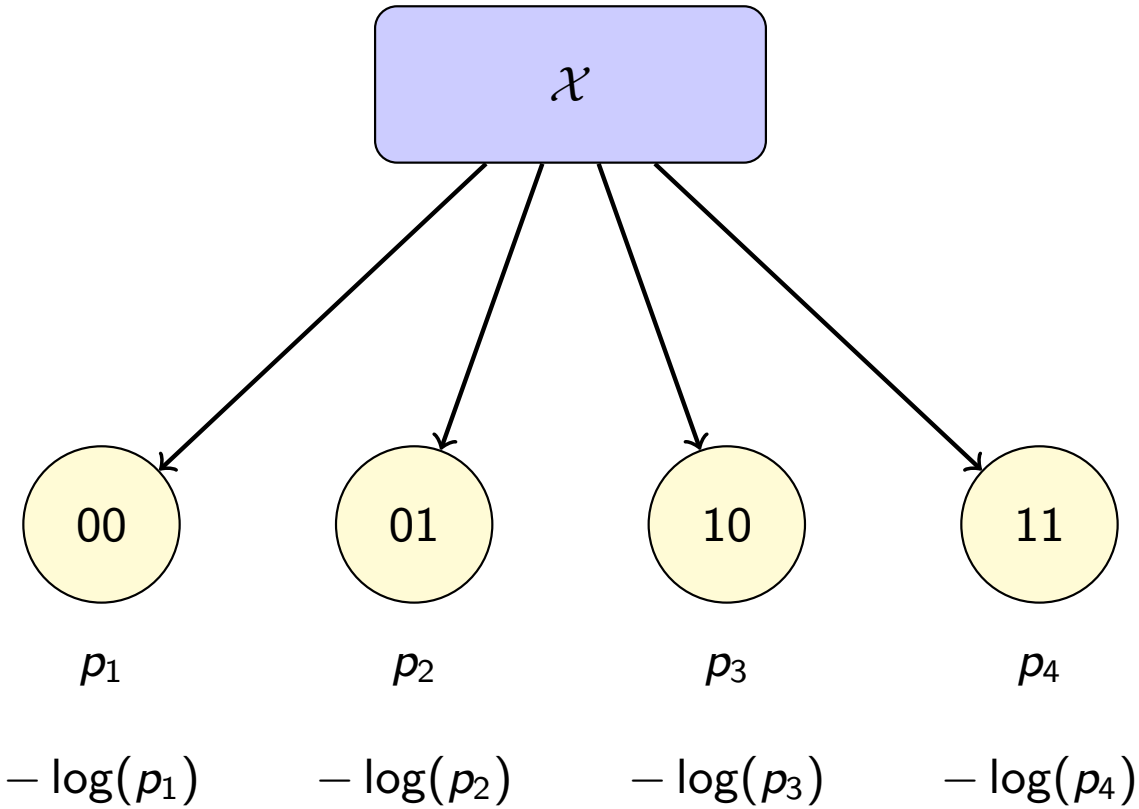
$$H(\mathcal{X}) = -\sum_{x \in \mathcal{X}} p(x) \log p(x),$$

where $0 \log 0$ is defined to be 0.

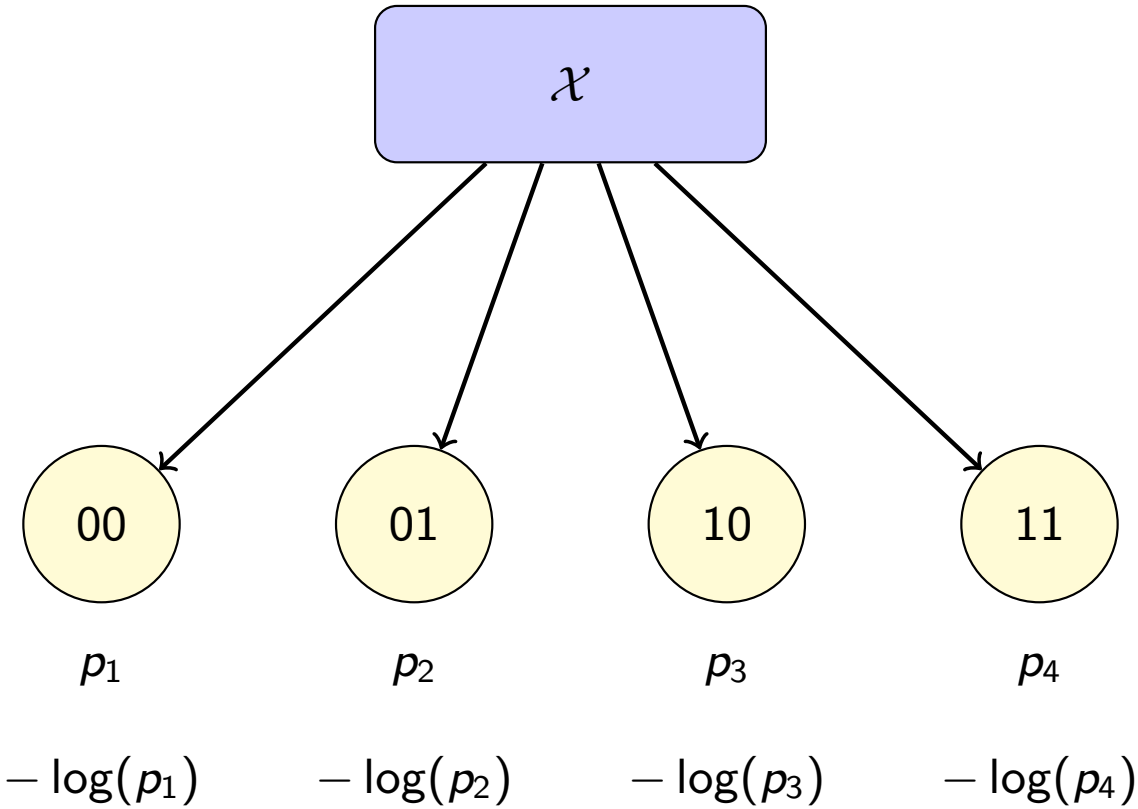
Shannon Entropy



Shannon Entropy

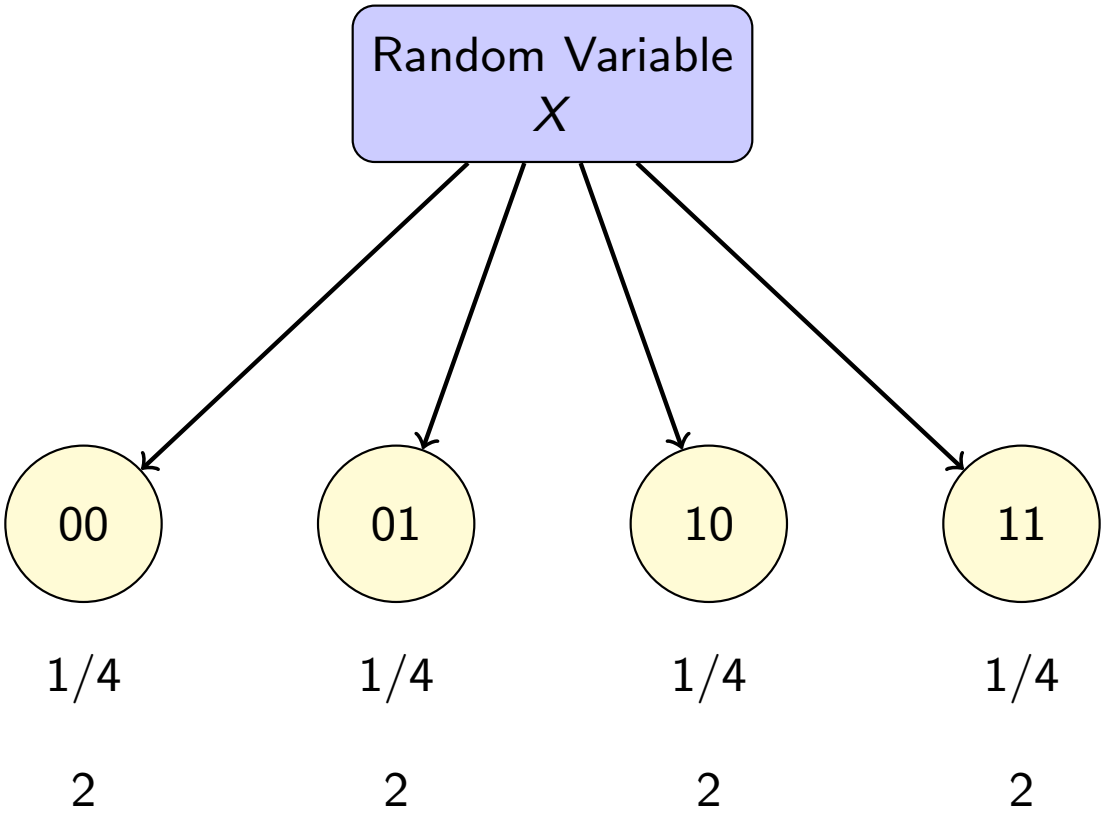


Shannon Entropy



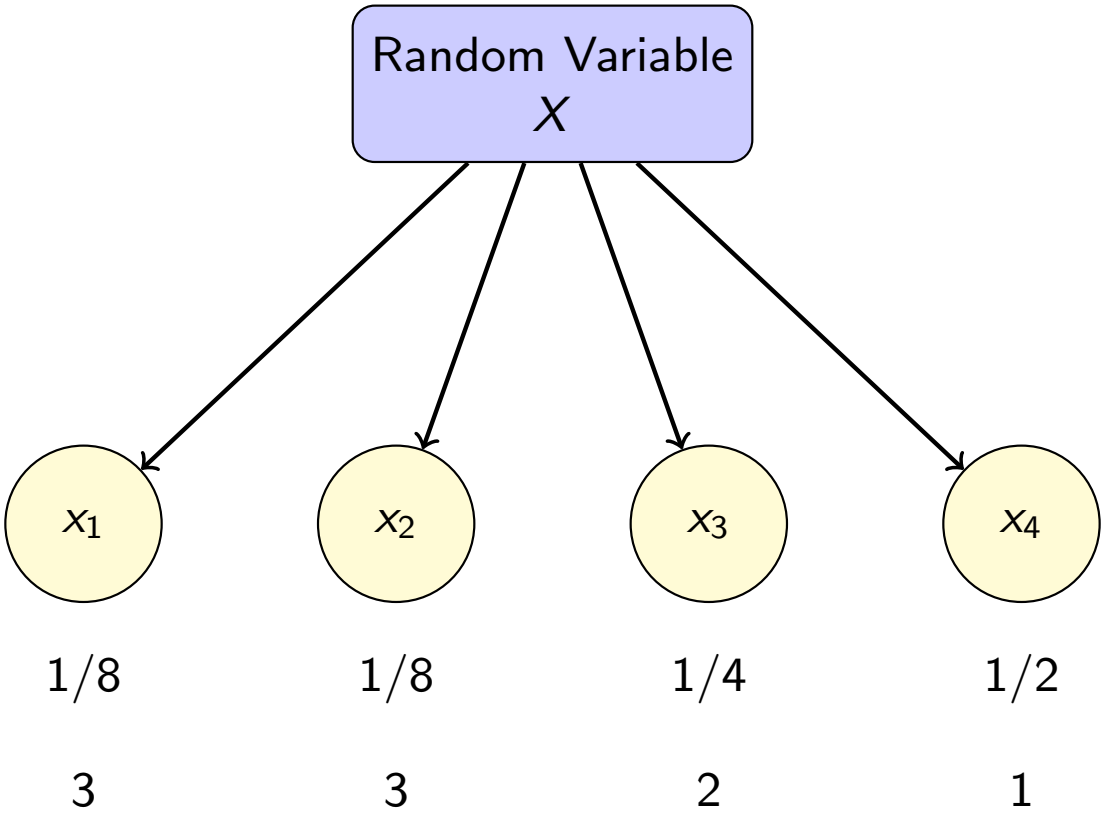
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Shannon Entropy



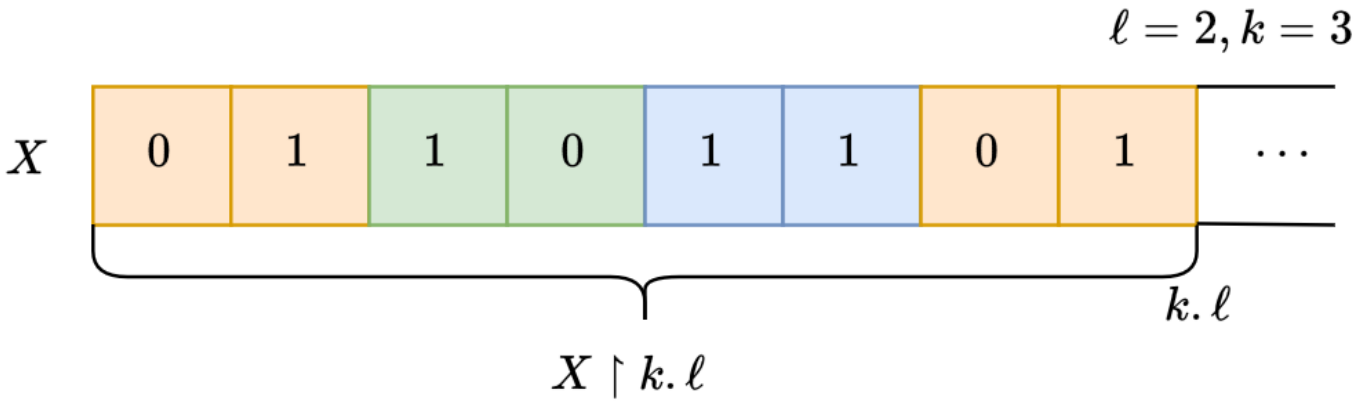
$$H(\mathcal{X}) = -\sum_{x \in \mathcal{X}} p(x) \log p(x) = 2$$

Shannon Entropy



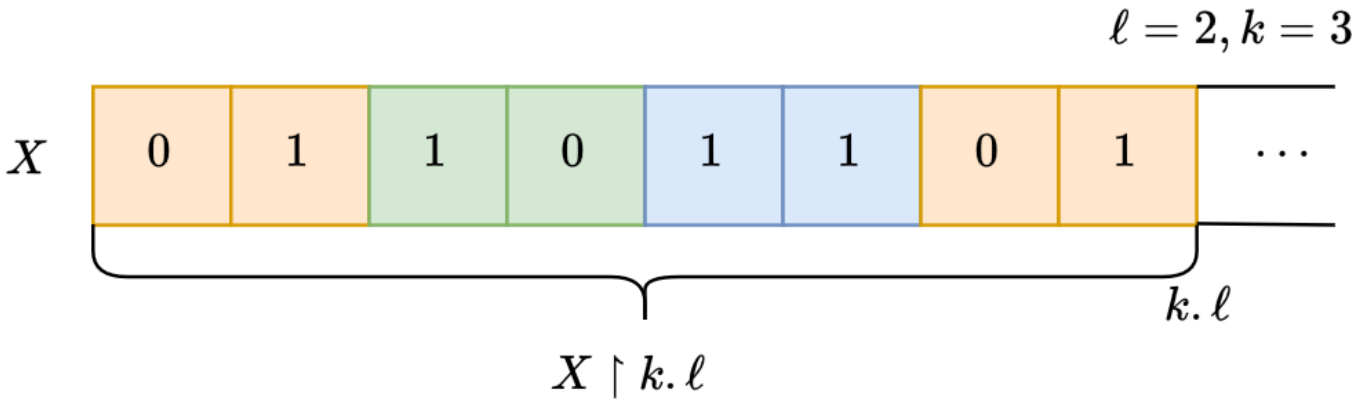
$$H(\mathcal{X}) = -\sum_{x \in \mathcal{X}} p(x) \log p(x) = 1.75$$

ℓ-length Disjoint Block Entropy



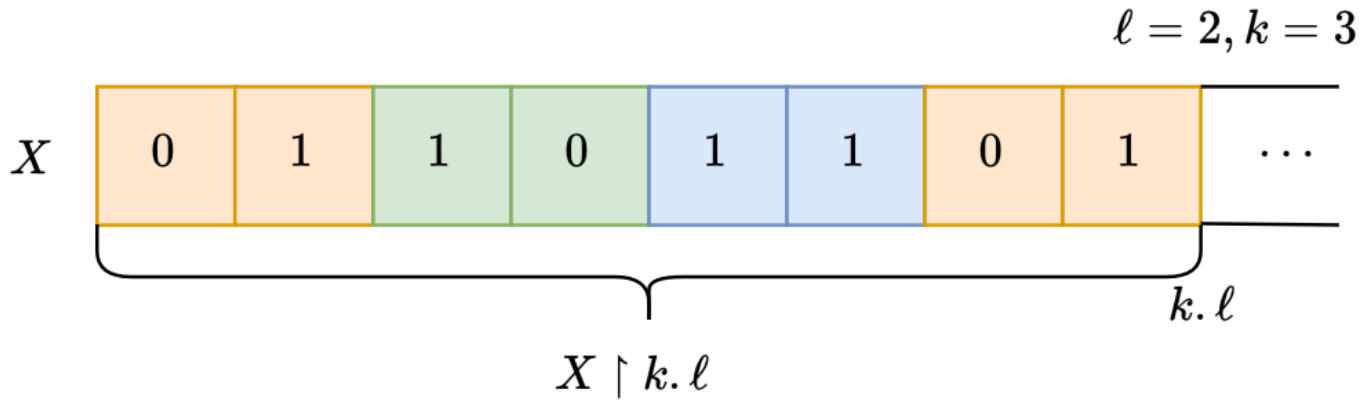
$$\#(00) = 0 \quad \#(01) = 2 \quad \#(10) = 1 \quad \#(11) = 1.$$

ℓ-length Disjoint Block Entropy



$$P(00) = 0/4 \quad P(01) = 2/4 \quad P(10) = 1/4 \quad P(11) = 1/4.$$

ℓ-length Disjoint Block Entropy



$$P(00) = 0/4 \quad P(01) = 2/4 \quad P(10) = 1/4 \quad P(11) = 1/4.$$

For $X \in \Sigma^\infty, \ell \in \mathbb{N}$, $k \in \mathbb{N}$,

$$P(u, X \upharpoonright k\ell) = \frac{\#(u, X \upharpoonright k\ell)}{k},$$

$\#(u, x)$: Number of *disjoint* occurrences of u in x.

ℓ-length Disjoint Block Entropy

Definition (ℓ-length Disjoint Block Entropy Rate)

The ℓ-length disjoint block entropy rate of $X \in \Sigma^\infty$:

$$H_\ell^{\text{disjoint}}(X) = \liminf_{k\to\infty} -\frac{1}{\ell} \sum_{u\in\Sigma^\ell} P(u, X \upharpoonright k\ell) \log P(u, X \upharpoonright k\ell).$$

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Disjoint Block Entropy

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Definition (Disjoint Block Entropy Rate)

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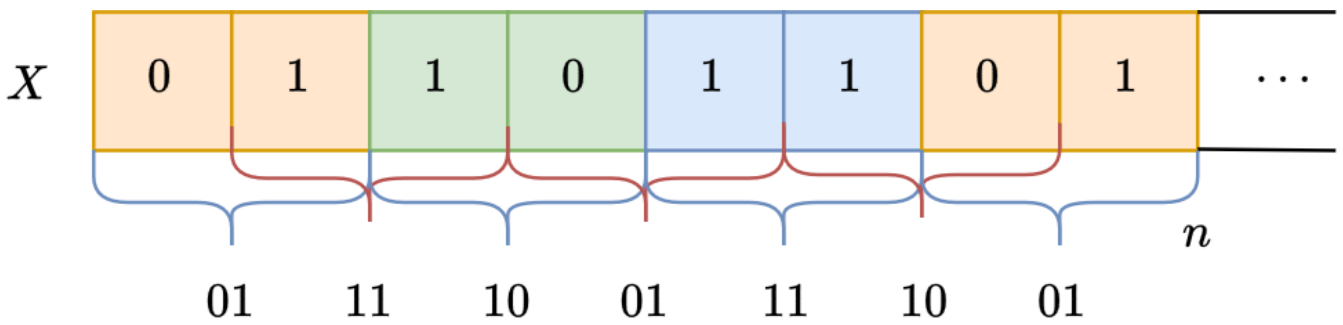
$$H^{\text{disjoint}}(X) = \inf_{\ell} H_{\ell}^{\text{disjoint}}(X).$$

ℓ-length Sliding Block Entropy

Let $x \in \Sigma^{\ell \cdot n}$ be a finite sequence and, for each block $u \in \Sigma^{\ell}$, define the frequency

$$P'(u,x) = \frac{N'(u,x)}{n-\ell+1},$$

where $N(u,x)$ is the number of sliding occurrences of u in x.



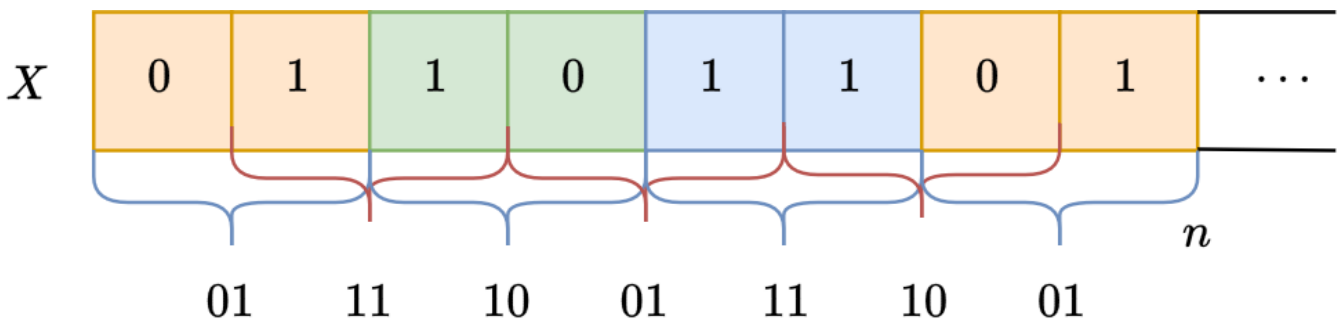
$$\#(00) = 0 \quad \#(01) = 3 \quad \#(10) = 2 \quad \#(11) = 2.$$

ℓ-length Sliding Block Entropy

Let $x \in \Sigma^{\ell \cdot n}$ be a finite sequence and, for each block $u \in \Sigma^{\ell}$, define the frequency

$$P'(u,x) = \frac{N'(u,x)}{n-\ell+1},$$

where $N(u,x)$ is the number of sliding occurrences of u in x.



$$P(00) = 0/7 \;\; \#(01) = 3/7 \;\; \#(10) = 2/7 \;\; \#(11) = 2/7.$$

Sliding Block Entropy

Definition (ℓ-length Sliding Block Entropy Rate)

The ℓ-length sliding block entropy rate of $X \in \Sigma^\infty$:

$$H_\ell^{\text{sliding}}(X) = \liminf_{k\to\infty} -\frac{1}{\ell} \sum_{u\in\Sigma^\ell} P(u, X \upharpoonright k\ell) \log P(u, X \upharpoonright k\ell).$$

Definition (Sliding Block Entropy Rate)

The disjoint block entropy rate of $X \in \Sigma^\infty$:

$$H^{\text{sliding}}(X) = \inf_\ell H_\ell^{\text{sliding}}(X).$$

Normality vs. Entropy

$$X \in \Sigma^{\infty} \text{ is Disjoint Normal } \iff H^{disjoint}(X) = 1.$$

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Theorem [Kozachinksy Shen 2021 [6]]

For all $X \in \Sigma^{\infty}$,

$$H^{\text{disjoint}}(X) = H^{\text{sliding}}(X)$$

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Entropy and Dimension

Theorem [Bourke, Hitchcock, Vinodchandran 2005 [5]]

For all $X \in \Sigma^\infty$,

$$H^{\text{disjoint}}(X) = \dim_{FS}(X).$$

Finite State gambler

Definition

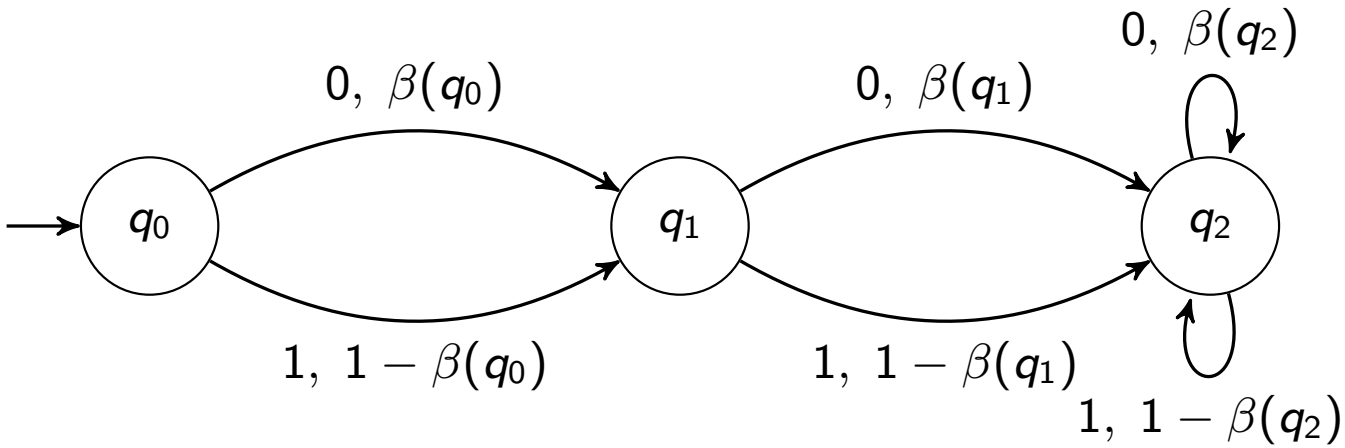
A finite state gambler is defined as

$$G = (Q, \delta, \beta, q_0, c_0),$$

where:

- Q is a finite set of states,
- $\delta : Q \times \{0, 1\} \to Q$ is the state transition function,
- $\beta : Q \to [0, 1]$ is the betting function,
- q_0 is the initial state, and
- c_0 is the initial capital.

Finite State gambler



Finite-state s-gales

Definition (Finite-state s-gale)

Let $s \in [0, \infty)$. A *finite state s-gale* is a function

$$d : \{0,1\}^* \to [0, \infty)$$

induced by a finite state gambler

$$G = (Q, \delta, \beta, q_0, c_0),$$

is defined recursively as $d^s_G(\lambda) = c_0$, and

$$d^s_G(w.b) = \begin{cases} 2^s \cdot d^s_G(w) \cdot \beta(q_w), & \text{if } b = 0 \\ 2^s \cdot d^s_G(w) \cdot (1 - \beta(q_w)) & \text{if } b = 1, \end{cases}$$

for every $w \in \{0,1\}^*$ and $b \in \{0,1\}$.

where $q_w = \delta^*(q_0, w)$ is the state reached after reading w.

Finite-state Dimension

Let $X \in \Sigma^\infty$ be an infinite sequence and let

$$S^\infty[d] = \{ X \in \Sigma^\infty : \limsup_{n \to \infty} d(X \upharpoonright n) = \infty \}.$$

And

$$\mathcal{G}_{FS}(X) = \{ s \in [0,1] : \exists \text{ a finite-state } s\text{-gale } d \text{ with } X \in S_\infty[d] \}.$$

Definition (Finite-state Dimension)

The *finite state dimension* of $X \in \Sigma^\infty$ is defined as

$$\dim_{FS}(X) = \inf \mathcal{G}_{FS}(X).$$

Entropy rates and Finite-state Dimension

Theorem (Bourke, Hitchcock, Vinodchandran)

For every sequence $X \in \Sigma^\infty$,

$$\dim_{FS}(X) = H^{disjoint}(X).$$

Outline

1 Overview

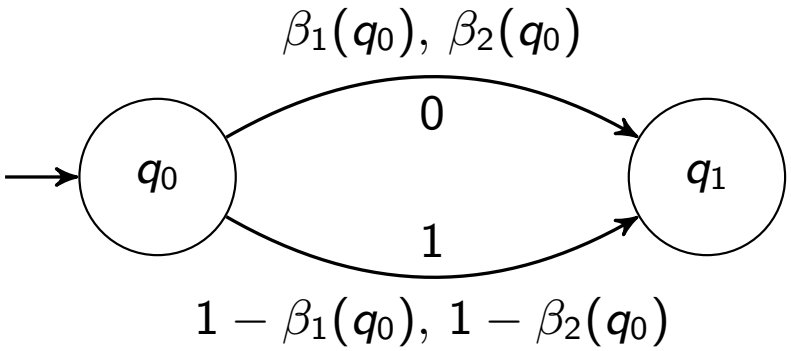
2 Normality

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Multi-Bet Finite State gambler



Multi-Bet Finite-State Gambler

Definition (Multi-Bet Finite-State Gambler)

A *multi-bet finite-state gambler* is a 5-tuple

$$G = (Q, \delta, \tilde{\beta}, q_0, c_0),$$

where:

- Q is a finite set of states.
- $\delta : Q \times \{0,1\} \to Q$ is the state transition function.
- $\tilde{\beta} : Q \to ([0,1]^k \cap \mathbb{Q}^k)$ assigns to each state $q \in Q$ a betting vector

$$\tilde{\beta}(q) = (\beta_1(q), \beta_2(q), \ldots, \beta_k(q))$$

  with $\sum_{i=1}^{k} \beta_i(q) = 1$.
- $q_0 \in Q$ is the initial state.
- c_0 is the initial capital.

s-gale Induced by a Multi-Bet Gambler

Let $G = (Q, \delta, \tilde{\beta}, q_0, c_0)$ be a k-bet finite-state gambler.

Definition (s-Product Gale induced by Multi-Bet FSG)

For a given $s \in [0, \infty)$, the *s-product gale* induced by G is the function

$$d_G^s : \{0,1\}^* \to [0, \infty)$$

defined recursively as follows:

$$d_G^s(\lambda) = c_0,$$

and for every $w \in \{0,1\}^*$ and $b \in \{0,1\}$,

$$d_G^s(w.b) = \begin{cases} d_G^s(w) \cdot \prod_{i=1}^{k} \left(2^s \cdot \beta_i(q_w)\right), & \text{if } b = 0, \\ d_G^s(w) \cdot \prod_{i=1}^{k} \left(2^s \cdot (1 - \beta_i(q_w))\right), & \text{if } b = 1. \end{cases}$$

where $q_w = \delta^*(q_0, w)$ is the state reached after reading w.

Multi-Bet Finite-State Dimension

Let $X \in \Sigma^\infty$ be an infinite sequence and let

$$\mathcal{G}_{\text{multi-bet}}(X) = \Big\{ s : \exists \text{ a multi-bet finite-state } s\text{-gale } d \text{ with } X \in S_\infty[d] \Big\},$$

where

$$S_\infty[d] = \Big\{ X \in \Sigma^\infty : \limsup_{n \to \infty} d(X \upharpoonright n) = \infty \Big\}.$$

Definition (Multi-Bet Finite-State Dimension)

The *multi-bet finite-state dimension* of X is defined as

$$\dim_{\text{FS}}^{\text{multi-bet}}(X) = \inf \mathcal{G}_{\text{multi-bet}}(X).$$

Our Results

Theorem

For every sequence $X \in \Sigma^\infty$,

$$\dim_{FS}^{\textit{multi-bet}}(X) = H^{\textit{sliding}}(X).$$

Proof Outline

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Theorem

For every sequence $X \in \Sigma^\infty$,

$$\dim_{FS}^{\textit{multi-bet}}(X) = H^{\textit{disjoint}}(X).$$

Our Results

Corollary

For every sequence $X \in \Sigma^\infty$,

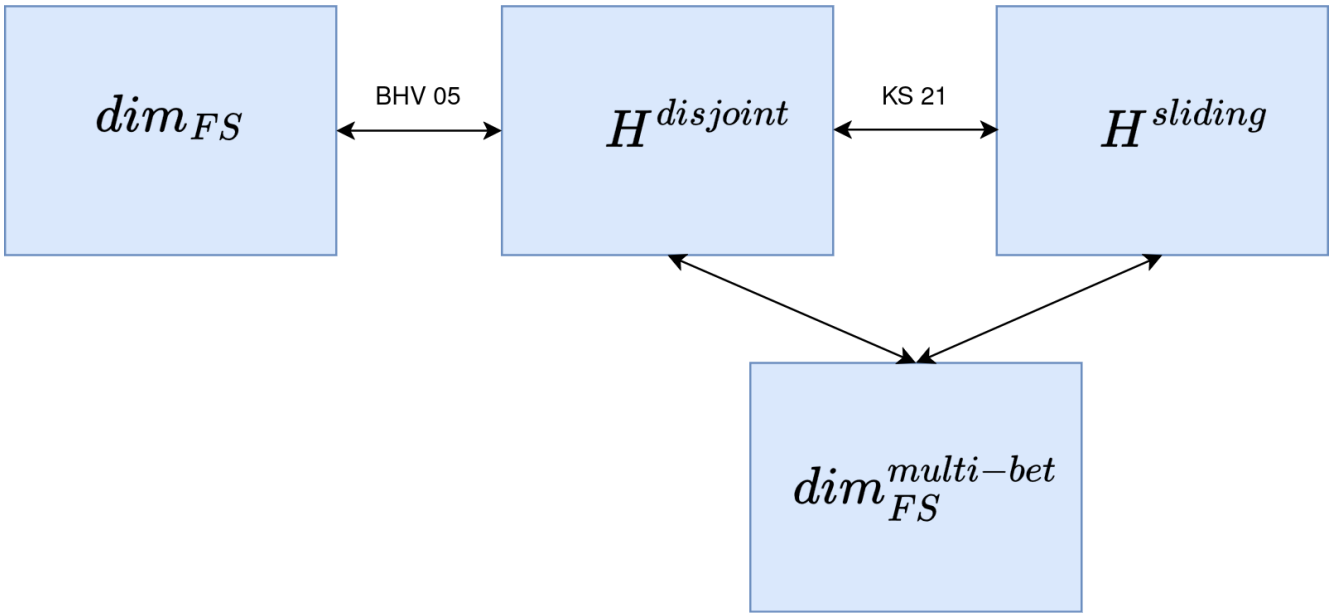
$$\dim_{FS}^{\textit{multi-bet}}(X) = \dim_{FS}(X).$$

Corollary

For every sequence $X \in \Sigma^\infty$,

$$H^{\textit{disjoint}}(X) = H^{\textit{sliding}}(X).$$

Conclusion



References


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C. Bourke, J. M. Hitchcock, N. V. Vinodchandran, *Entropy rates and finite-state dimension*, Theor. Comput. Sci., 2005.


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Thank you!

Proof Ideas

Lemma

For every sequence $X \in \Sigma^\infty$,

$$\dim_{FS}^{\textit{multi-bet}}(X) \leq H^{\textit{sliding}}(X).$$

Proof Idea

We show:

$$H^{\textit{sliding}}(X) < s \implies \exists\, \ell\text{-bet } s\text{-gale that succeeds on } X$$
$$\implies \dim_{FS}^{\text{multi-bet}}(X) \leq s.$$

Proof Ideas

Lemma

For every sequence $X \in \Sigma^\infty$,

$$\dim_{FS}^{\textit{multi-bet}}(X) \leq H^{\textit{sliding}}(X).$$

Proof Idea

$$\begin{aligned} H^{\textit{sliding}}(X) < s &\implies \inf_\ell H_\ell^{\text{sliding}}(X) < s \\ &\implies \exists \ell \; H_\ell^{\text{sliding}}(X) < s \\ &\implies \exists \ell \; \liminf_{k\to\infty} -\sum_{u\in\Sigma^\ell} P(u, X \upharpoonright k\ell) \log P(u, X \upharpoonright k\ell) < s\ell. \end{aligned}$$

Assume $\forall u \in \Sigma^\ell$, $\lim_{k\to\infty} = P(u, X \upharpoonright k\ell) = \mathbb{P}(u)$

Proof Ideas

Lemma

For every sequence $X \in \Sigma^\infty$,

$$\dim_{FS}^{\textit{multi-bet}}(X) \leq H^{\textit{sliding}}(X).$$

Proof Idea

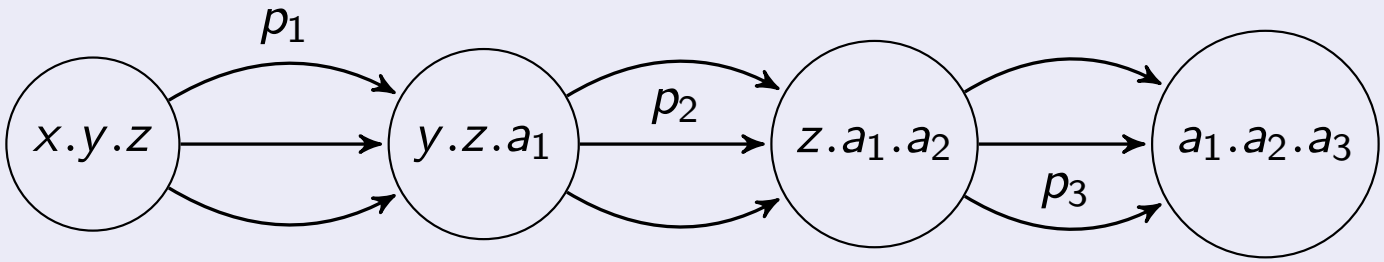
Assume $\forall u \in \Sigma^\ell$, $\lim_{k\to\infty} = P(u, X \upharpoonright k\ell) = \mathbb{P}(u)$

We show that there exists an ℓ-bet s-gale G_ℓ such that:

- G_ℓ places cumulative bet of $\mathbb{P}(u)$ on every sliding occurance of u.
- $H(\mathbb{P}) < s \cdot \ell \implies G_\ell$ succeeds on X.

Proof Idea (contd)

$H(\{\mathbb{P}(w)\}_{w\in\Sigma^\ell}) \leq s\ell.$

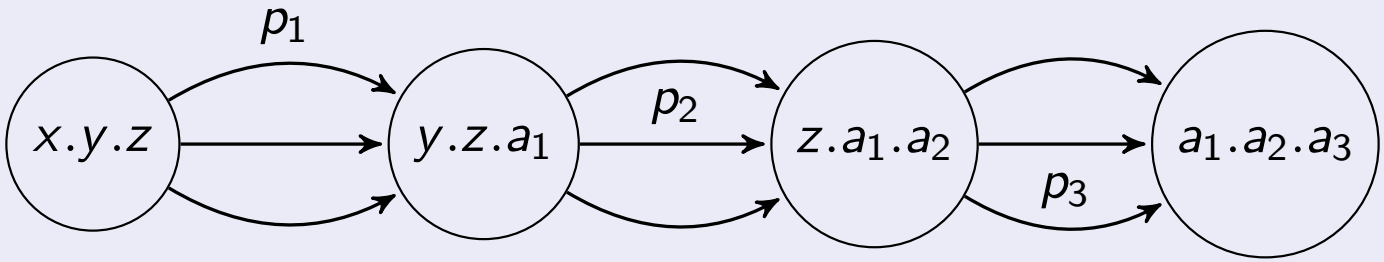


$p_1 = \mathbb{P}(a_1), p_2 = \mathbb{P}(a_2|a_1), p_3 = \mathbb{P}(a_3|a_1, a_2).$

$$p_1 \cdot p_2 \cdot p_3 = \mathbb{P}(a_1.a_2.a_3).$$

Proof Idea (contd)

$H(\{\mathbb{P}(w)\}_{w\in\Sigma^\ell}) \leq s\ell.$



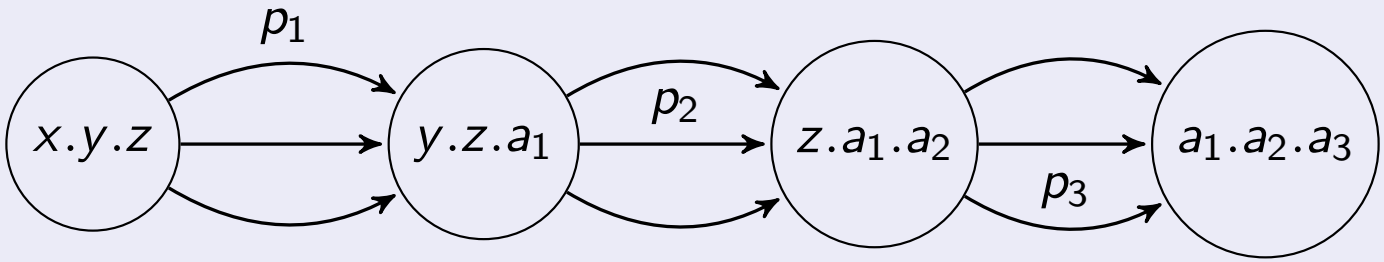
$p_1 = \mathbb{P}(a_1), p_2 = \mathbb{P}(a_2|a_1), p_3 = \mathbb{P}(a_3|a_1, a_2).$

$$p_1 \cdot p_2 \cdot p_3 = \mathbb{P}(a_1.a_2.a_3).$$

$d^s_{G_\mathbb{P}}(X \upharpoonright n-\ell) \geq \prod_{w_i\in\Sigma^\ell}(2^{s.\ell} \cdot \mathbb{P}(w_i))^{n_i}.$ $\qquad$ $n_i = \#_{w_i}(X \upharpoonright n-\ell).$

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$$\log(d^s_{G_\mathbb{P}}(X \upharpoonright n-\ell)) \geq n-\ell \left(s\ell - \sum_{w_i\in\{0,1\}^l} P(w_i, X \upharpoonright n-\ell)\ \log(1/\mathbb{P}(w_i)) \right).$$

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